NOVEANTO NUR AKBAR

AI & Data Engineer | Applied Machine Learning & Core Banking Data Analytics

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PROFESSIONAL SUMMARY

Informatics graduate (**S.Kom, GPA 3.45/4.00**) from UPN "Veteran" Yogyakarta specializing in AI engineering, predictive data modeling (Python, SQL), and secure enterprise data infrastructure. Proven track record at berijalan (Astra Credit Companies) modernizing enterprise AI architectures, engineering two-tier caching pipelines that **reduced monthly inference costs by up to 61%**, and enforcing zero-trust multi-tenant data boundaries. Deep learning research in time-series stock forecasting (**MAPE 2.29%, RMSE 3.13**) and a privacy-preserving banking Text-to-SQL data mart copilot compliant with UU PDP No. 27/2022. Cross-functional leadership coordinating **28 universities**. Eager to apply data analytics, operational discipline, and scalable AI to optimize banking business processes.

EDUCATION

Universitas Pembangunan Nasional "Veteran" Yogyakarta 2022 – Mar 2026

Bachelor of Informatics (S.Kom) | GPA: 3.45 / 4.00 | Yogyakarta, Indonesia

- **Undergraduate Thesis:** "Analisis Pengaruh Fitur Teknikal terhadap Akurasi Prediksi Harga Saham NVIDIA Menggunakan Bi-LSTM" – benchmarked 10 neural network variants with RSI, MACD, and Bollinger Bands, achieving **2.29% MAPE** and **3.13 RMSE**.
- **Relevant coursework:** Database Systems, Data Structures & Algorithms, Machine Learning, Cloud Computing, Software Engineering, Applied Statistics.

PROFESSIONAL EXPERIENCE

Fullstack Developer Jul 2026 – Present

berijalan · PT Cipta Sedaya Digital Indonesia (CSDI) | Yogyakarta, Indonesia | Bootcamp Batch 18 (Jul – Oct 2026), Freelance (Oct 2026 – Present)

- **Full-Stack AI Workplace Modernization:** Decoupled monolithic Laravel services into an Express.js and Next.js 16 architecture for Anythink (Astra Credit Companies), enabling async document ingestion and low-latency SSE token streaming.
- **Inference Cost Optimization:** Architected a Two-Tier FinOps Gateway combining local in-memory BGE-M3 semantic caching (sub-30ms CPU inference) and Microsoft LLMLingua-2 context pruning, eliminating 100% API costs on semantic hits and **cutting monthly inference spend by ~61%**.
- **Multi-Tenant Data Governance:** Enforced schema validation and sanitized OData query filtering in Azure AI Search, guaranteeing **100% multi-tenant data isolation** with no cross-tenant leaks.
- **Cryptographic Backward-Compatibility:** Engineered a timing-safe AES-256-CBC/HMAC-SHA256 Node.js decryption bridge for legacy PHP data, achieving zero-data-loss, zero-downtime migration.
- **Engineering Recognition:** Awarded "Best Progressive Developer" in Bootcamp Batch 18 for delivering production-ready enterprise solutions.

Co-Founder & Software Engineer May 2026 – Present

KickOff Studio | Yogyakarta, Indonesia

- **E-Commerce Web Architecture:** Directed full-stack development for an on-demand merchandise platform (Next.js App Router, Tailwind CSS) with interactive preview customizers.
- **Billing & Operations Automation:** Built a centralized administrative back-office portal integrated with Google Sheets API, automating invoice and receipt (nota) generation to **eliminate manual billing friction by 80%**.
- **Communication Infrastructure:** Configured Cloudflare Email Routing for @kickoffstudio.online with centralized mailboxes at zero server overhead.

AI Trainer – Multimodal & Reasoning Jul 2026 – Present

Outlier.ai | Remote, United States (Freelance)

- **Quality Calibration & RLHF:** Executed pairwise evaluations and adversarial stress tests on frontier voice and multimodal AI models to optimize alignment datasets.
- **Logic & Data Verification:** Audited instruction following, multi-step math reasoning, and hallucination containment to ensure factual accuracy.

Frontend Developer Intern Feb 2025 – Apr 2025

LPPM UPN "Veteran" Yogyakarta | Yogyakarta, Indonesia

- **Campus-Wide Workflow Digitization:** Built the administrative web interface and MySQL pipelines (PHP, JavaScript), replacing manual paper filing with a centralized validation system.
- **Institutional Adoption:** Rolled out the web-based ISBN submission system adopted university-wide by dosen, students, and tendik across all faculties, resolving review bottlenecks for **300+ submissions**.

Media Operations Staff (Volunteer) Nov 2023 – Nov 2025

FIFA & PSSI | Surakarta & Sleman, Indonesia

- **International Event Coordination:** Managed pitchside media operations across **5 international football tournaments**, including the FIFA U-17 World Cup 2023 and ASEAN Championships.
- **Regulatory Compliance & SLA Delivery:** Enforced FIFA/AFF media access protocols, coordinated mixed-zone interviews, and delivered broadcast assets to team analysts under strict deadlines.

KEY TECHNICAL PROJECTS

Sentra-SafeQuery: Privacy-Preserving Banking Text-to-SQL Copilot 2026

Python, Streamlit, SQLGlot, SQLAlchemy, PostgreSQL

- Engineered a Text-to-SQL copilot enabling frontline branch officers to explore core banking data marts in natural Indonesian.
- Designed an in-memory ephemeral SessionVault replacing customer PII (16-digit NIK, account numbers, names) with tokenized tags ([NIK_01]), achieving **100% compliance** with UU PDP No. 27/2022.
- Built an AST SQL Firewall (SQLGlot) enforcing root SELECT assertions, blocking CTE/DML mutations via recursive node traversal, blacklisting system functions, and capping rows (LIMIT <= 100).
- Integrated an Indonesian Banking Semantic Dictionary mapping regional operational idioms (savings, agent banking, QRIS, mobile banking, micro-lending) to relational dimensional constraints.

Bandha: Multi-Modal AI Personal Finance Platform 2026

Flutter, Express.js, Groq Llama 3.3, Gemini 2.5, Google Sheets API

- Built a cross-platform expense app and Telegram bot with a Quad-AI routing pipeline: Groq Whisper Large v3 (voice STT), Llama 3.3 (sub-450ms entity extraction), Gemini 2.5 Flash (receipt OCR), and GPT-OSS 120B (financial analysis).
- Engineered a serverless ledger on Google Sheets API (Service Account JWT), **reducing hosting costs to $0.00** with full user data sovereignty.
- Implemented an in-memory slot-filling state machine with dynamic missing-attribute prompting and /undo transaction rollbacks.

ORGANIZATIONAL & LEADERSHIP EXPERIENCE

Head of Talent & Interest Division Feb 2025 – Feb 2026

HIMATIF UPN "Veteran" Yogyakarta

- Steering Committee for **3 flagship programs**: Veternity (national tech competition with **28 universities**), IF Cup (campus-wide tournament), and IF Corner.
- Supervised cross-functional divisions, enforcing operational timelines, milestone governance, and budget allocations.

Media Staff & Head of Media, IIO 2024 Feb 2024 – Feb 2025

BEM FTI UPN "Veteran" Yogyakarta

- Standardized digital publication workflows for **2,000+ engineering students** across official BEM FTI channels.
- Led the media division for Indonesian Industrial Outlook 2024: campaign marketing, live coverage, attendee data tracking.

CERTIFICATIONS & HONORS

- **Junior Mobile Programmer** – KOMDIGI (Kominfo VSGA), Credential ID: 193107451110-11 (Valid: Aug 2025 – Aug 2028).
- **Best Progressive Developer** – Fullstack Developer Bootcamp Batch 18, berijalan (Oct 2026).
- **Co-Author, Published Paper** – RSF Conference Series (ICSRD-2025): "Web-Based ISBN Registration System at LPPM UPN 'Veteran' Yogyakarta Using the RAD Method" (2025).
- **English Proficiency** – Institutional TOEFL Score: 520.

TECHNICAL SKILLS

Data Analytics & Modeling: Python (Pandas, NumPy, Scikit-learn, TensorFlow), SQL (PostgreSQL, MySQL, SQLite), Time-Series Forecasting (Bi-LSTM), Statistical Modeling, Metric Evaluation (MAPE, RMSE), Data Cleaning, Data Visualization (Plotly, Matplotlib)

AI & Backend Engineering: LLM / RAG, SQLGlot (AST), LangChain / LangGraph, Redis Stack (Vector HNSW), Semantic Caching, Prompt Compression (LLMLingua), Next.js, Express.js, Node.js, Prisma ORM, RESTful APIs, SSE

Banking & Compliance Governance: UU PDP No. 27/2022, PII Masking, Tokenization, Zero-Trust Architecture, Core Banking Terminology (savings, agent banking, QRIS, mobile banking)

Tools & Platforms: Docker, Git, Google Sheets API, AWS EC2, Cloudflare, Streamlit, Postman, Linux / WSL2

Soft Skills & Languages: Executive Reporting, Cross-Unit Coordination, Analytical Problem Solving; Indonesian (Native), English (Fluent, TOEFL 520)